Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/16/2014
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Plant Mechanical Signaling During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	12/31/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2015 per NSSC information (Ed., 1/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Swanson, Sarah (University of Wisconsin-Madison)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants are exquisitely sensitive to mechanical stimulation which triggers changes ranging from widespread alteration in patterns of gene expression to profound shifts in development. Indeed, so pervasive is the mechanical stimulation from internal and external forces, such as the weight of tissues and organs, to a biology that has evolved under a constant 1xG, that the resultant signaling is thought to provide a required regulatory background to control a wide range of plant developmental and physiological processes. Despite this acknowledged role in plant morphogenesis, the overall contribution of such mechanical cues to plant function remains poorly defined, largely due to the difficulties in removing the pervasive 1xG-driven mechanical signal when conducting experiments on Earth. The microgravity environment of the International Space Station (ISS) therefore provides a unique tool to both directly address this hypothesis of mechanical load-driven morphogenesis, as well as to ask what features of plant growth in space reflect disruption of this mechano-signaling regulatory network. We therefore propose to grow seedlings of Arabidopsis thaliana on board the ISS using the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware. We will then compare their patterns of growth and gene expression to ground-based controls to ask how the well-defined "fingerprints" of mechanical response are altered in space. In addition, we plan to characterize plants grown in space whose mechanical signaling is artificially activated or inhibited by mutations in a gene (TCH2) which has an established role in the mechanical signaling system of plants. Results from such analyses are responsive to the goals of the NASA Research Announcement in addressing the fundamental question of the role of mechanical loading on plant growth and development on Earth as well as probing how manipulating this system through targeting specific mechanoresponse genes might be used to tailor plant growth to thrive in the unique environment of spaceflight. This experiment will also be supplemented with a parallel geneLAB experiment to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. These variants (ecotypes) have been chosen to span the range of likely ecotypes to be used in future spaceflight experimentation and so will provide key baseline comparisons to help understand the impact of ecotypic makeup on responses to spaceflight. This element of the research will therefore provide important background data for the design of future spaceflight experimentation using this widely adopted model plant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how mechanical sensing operates in plants using Arabidopsis thaliana (Mouse Ear Cress). The work targets the Ca2+-dependent signaling machinery and a specific element of this machinery, a gene named touch-2, also known as calmodulin-like 24 (cml24). Wild type and mutants in cml24 will be grown on the International Space Station (ISS) and their patterns of growth and gene expression compared to paired plants grown on Earth. The results from this work will not only provide insight into how the reduced gravity environment of spaceflight affects plant growth but also has impact on understanding similar responses on Earth. On Earth, mechanical signaling systems are known to affect plant vigor, alter their growth habit (favoring plants that are smaller and more robust) and also impacts on plant defense responses to pathogens. Therefore, this work aimed at analysis of how Ca2+ signaling is involved in the altered mechanical responses induced during spaceflight should also reveal the role for this system in the more general mechanically-linked responses of plants on Earth. Importantly, this research will allow analysis of the effects of the plant's own weight on its growth and development, research only possible in the microgravity environment of spaceflight. These results should provide molecular targets for potential manipulation to tailor plant growth to varied mechanical environments, especially the relatively mechanically silent environments of spaceflight and controlled enclosed environments on Earth, such as greenhouses.
Task Progress:	Both the BRIC19 touch-related experimentation and the geneLAB ecotype projects have completed their Science and Payload Verification tests. Preflight testing: The setup of the both the BRIC19 and geneLAB samples capitalized on the successful design used for the BRIC17 experiment flown in 2013. Surface sterilized seeds were plated on nutrient agar medium, assembled into Petri Dish Fixation Units (PDFUs), and loaded into the BRIC hardware. The BRIC was then held at 4°C to delay germination over the 4 days required to mimic prelaunch, launch, and transit to the International Space Station (ISS). The BRIC was then warmed to ISS ambient temperature, seeds allowed to germinate, and after 8 days these were fixed by injection of RNAlater. The samples were then frozen at -80°C and after 2 weeks storage were thawed and sampled. Testing showed that germination was reliably delayed by holding the plates at 4°C until the required time for seedling growth. These verification tests also showed >90% germination of seeds in the BRIC-PDFU hardware upon warming to room temperature. One exception was the Ws ecotype from the geneLAB samples (see below). Analysis of images of the germinated seedlings showed they had reached the expected size from previous ground-based studies for all seed lines tested. RNA isolation from these samples also showed yields of >1.5 µg/sample with a quality exceeding the requirements for analysis by either RNAseq or PCR, the two gene expression techniques to be used on the spaceflight samples. When testing the geneLAB samples, 3 ecotypes were analyzed (Landsberg, CVi, and Ws). Of these Ws showed 0% germination during initial testing. Troubleshooting identified an absolute light requirement for germination in this ecotype that was not supplied in the sealed, dark interior of the BRIC hardware. Therefore, a protocol was developed where seeds were planted onto nutrient agar, irradiated with white light at 70 µMoles m-2 s-1 for 12 h, and then loaded into the PDFUs and treated as described previously. Testing showed this light treatment promoted >90% germination in Ws and did not alter the effectiveness of the subsequent storage at 4°C in inhibiting germination for >4 days. A full Payload Verification Test (PVT) is under way to assess the effectiveness of this approach or whether an alternate ecotype to Ws will need to be flown. If the light pretreatment proves effective in this PVT all ecotypes will be pretreated with 12 h light for the flight experiment. Ground-based analysis: In parallel to the flight experiment, wild-type, and the cml24 touch mutants to be used in the spaceflight experiment have been monitored for their responses to touch and gravitropic stimulation on Earth. Plants expressing the GFP-based 'Yellow Cameleon Nano-65' Ca2+ sensor were generated and Ca2+ changes in response to touch and gravitropic stimulation (90° rotation) monitored by confocal ratio imaging. cml24 mutants show altered touch-induced growth and effects on Ca2+ signals are currently being assessed.

Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Jayaraman D, Gilroy S, Ané JM. "Staying in touch: mechanical signals in plant-microbe interactions." Current Opinion in Plant Biology. 2014 Aug;20:104-9. Epub 2014 May 27. Review. http://dx.doi.org/10.1016/j.pbi.2014.05.003 ; PubMed PMID: 24875767 (NOTE: reported originally as "in press, as of May 2014") , Aug-2014